

Dual ordinal spectral ranking of journals and institutions

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Abstract

We develop a dual ordinal spectral ranking of journal and institution units that determines influence-per-work ranking scores via block citation matrices. There are four bibliometrically distinct structures connecting journals (J) and institutions (I): J to J (similar to the Clarivate Eigenvector score), I to I (the institutional counterpart), J+I to J+I (full 2-unit citation matrix), and JI to IJ (recirculating). Applied to the economics and commerce journal literature using OpenAlex data, the method produces stable, interpretable joint rankings whose sensitivity to model parameters is examined systematically. Journal rankings are stabilised by the inclusion of institutions, and both unit's rankings remain remarkably similar (Spearman score ρ 0.99) over the period 2000-2024.

Keywords: spectral ranking, dual journal-institution prestige, Katz-Hubbell resolvent

1 Introduction

The research system relies on journals and institutions. Journals filter works through editorial and peer review, archive the literature, and create metadata for retrieval and exploration. Institutions fund individual researchers, implement training, oversee compliance, establish research organisational units and maintain physical and information infrastructure. Although the status hierarchy that links journals and institutions is a pervasive aspect of the research system (Merton, 1973; McNamee and Willis, 1994; Martin, 1998), bibliometric studies that explore the joint hierarchy of journals and institutions are infrequent. This paper proposes a joint ordinal spectral ranking of journals and institutions derived from bibliometric data. The method helps to reveal how status is formed and maintained. We intend to apply the approach to study the relations between the status of journals and institutions, and researcher esteem in later work.

The introduction of the science citation index (Garfield, 1964) opened a new way to investigate connections between journals and institutions. Citation-based journal ranking in particular became a useful means to manage collections and to select outlets for publications. On the other hand, the misapplication of the journal impact factor (JIF) as proxy for the importance of research works published in that journal fostered a still-unresolved tension in bibliometric evaluation (Garfield, 1979; MacRoberts and MacRoberts, 2018). Perhaps due to the complexity of their disciplines and missions, an institution impact factor (IIF) has not emerged although the Scimago Institution and. Rather, institutional rankings derived from bibliometric indicators of works, authors or journals are often composed with other indicators to form an institutional score. This is the approach in global rankings such as QS, Times Higher Education (THE), Shanghai (ARWU), Leiden Ranking (CWTS) and SCImago Institutions Rankings, as well as many national performance-based research funding schemes (Zacharewicz et al, 2019).

The prospects for using a citation index to explore research importance were considered in an early work by Narin (1976), Pinski and Narin (1976), and Narin et al (1976) who proposed a recursive method to construct influence weights of journals within a disciplinary field. To obtain influence weights for institutions, Narin summed the institutional publications in a field weighted by the respective journal influence score. Over a sample of several dozen universities, the total institution influence scores were well-correlated with survey-based rankings, but also with the number of publications, suggesting that survey respondents combined influence and size. Narin's work built on earlier studies of social networks and developed into the eigenvector approach now widely used for journal ranking (Bergstrom et al, 2008; Vigna, 2016; Franceschet, 2010). Eigenvector methods have also been applied to author-level influence weightings in the field of social science, and extended by Narin's approach to ranking institutions and countries (West et al, 2013). An alternative approach iterates between the article-article network and the corresponding article-author network (Ujum

et al, 2015; Pal and Ruj, 2015). The formulation by Cao et al (2023) adopts a similar network, with reference-based article-article links, and the related article-institution links, ranked by a novel and effective iteration over random walks. This is one of the few spectral rankings of institutions. The authors note that a limiting case of this approach corresponds to the eigenvector (PageRank) method: Geller (1978) noted that random walks over Markov chains are closely related to the spectral ranking introduced by Narin.

We foreground for readers some analytical devices adopted in this paper. Firstly, this paper views references and citations through the lens of Social Systems Citation Theory (Tahamtan and Bornmann, 2022, SSCT). It concentrates on the structural features of the network of references and citations, setting aside authorial, editorial and managerial intent. This approach contrasts with actor-network theory (Latour, 2007, ANT), which foregrounds the intentful roles of heterogeneous agents including authors in assembling citation networks. ANT will be important in our follow-up studies that relate this paper to the referencing activities of individual researchers. Second, we say that work w_1 *references* a work w_2 in the reference list of work w_1 , and that w_2 is *cited by* w_1 (Wouters, 1999). We say that a work w_1 pays or gives *attention* to its references, and that the references *influence* the work. We explain below how attention and influence at the work level are projected to journals and institutions.

The paper proceeds as follows. In the next section, we construct a framework for computing the transfer of attention or influence among journal and institution units coupled by the reference lists of the works they publish. Numerical methods for computing these ordinal spectral rankings are presented and their convergence discussed. We then construct a corpus of works in the fields of Economics and Business, forming subsets of the corpus to illustrate the interconnections of units, fields, and time. We present spectral rankings over these subsets, alongside sensitivity and bootstrap studies of some key parameters. The paper discusses these results including a critique of our methods, and concludes with some directions for future work.

2 Mathematical framework: dual spectral ranking

We begin by selecting N_s sources $\mathfrak{S}$ in a set of fields, disciplines or categories $\mathfrak{F}$. Then we select N_w works $\mathfrak{W}$ published in $\mathfrak{S}$ within the census window years $t_c = [t_0^c, t_1^c]$ or target years $t_t = [t_0^t, t_1^t]$. From the reference lists of the set of census works $\mathfrak{W}^c$ we identify the set of target works $\mathfrak{W}^t$ and form the filtered reference set $\mathfrak{R} = \{r(i, j) : w_i \in \mathfrak{W}^c, w_j \in \mathfrak{W}^t\}$. Author affiliations in $\mathfrak{W}$ establish a set of institutions $\mathfrak{U}'$ and authors $\mathfrak{A}'$. In practice many institutions (and sources) publishing only a few works per year can be dropped from the corpus. Let $\bar{w}_u$ be the mean annual census-window work count for institution u and $\bar{w}_s$ the corresponding work count for source s . We determine lower limits τ_U and τ_S by examining the respective frequency distributions and define $\mathfrak{U} \subseteq \mathfrak{U}'$ as the institutions with $\bar{w}_u \geq \tau_U$, and $\mathfrak{S} \subseteq \mathfrak{S}'$ as the sources with $\bar{w}_s \geq \tau_S$, with corresponding author set $\mathfrak{A} \subseteq \mathfrak{A}'$. Works that lose all their authorships under the institution filter, or whose source is dropped by the source filter, are removed from the corpus. The choice to apply τ_U and τ_S once across the corpus or else within each time window is examined below.

Let $R_i = |\{j : (i, j) \in \mathfrak{R}\}|$ be the number of references from work w_i that land in the filtered set $\mathfrak{R}$, and $\bar{R} = |\mathfrak{R}|/N_w$ the mean of R_i over $\mathfrak{W}$. Since filtered reference lists have different lengths, we consider two cases for weighting references,

$$\rho_i = \begin{cases} 1 & \text{(full reference counting)} \\ \bar{R}/R_i & \text{(fractional reference counting)}. \end{cases} \quad (1)$$

Under full reference counting a work with twice as many filtered references has twice the prominence; under fractional reference counting every work's filtered references count equally (Zitt and Small, 2008).

A work has a retained source and one or more institutions. For journal spectral ranking, it is straightforward to link the work to its journal. In dual spectral ranking, however, we should consider (i) the fractionation between the journal and the institutions, and (ii) the fractionation across institutions. The parameter χ fractionates the source (χ) and institutions ($1 - \chi$); it balances the two unit types by population when $\chi = N_s/(N_s + N_u)$, which we adopt as the natural choice χ^* for the full-joint mask in Section 4.

There are two models fractionating a work’s institutions (i.e. the *institution weight*)

$$\omega_{iu} = \begin{cases} \sum_{\ell: u \in \mathcal{U}_{i\ell}} \frac{1}{a_i} \frac{1}{u_{i\ell}}, & \text{(author-fractional)} \\ \frac{1}{n_i}, & \text{(institution-fractional).} \end{cases} \quad (2)$$

Here, $\mathcal{U}_{i\ell}$ is the set of institutions of author ℓ of work w_i , $u_{i\ell} = |\mathcal{U}_{i\ell}|$ is its cardinality, a_i is the author count of work w_i , and $n_i = |\bigcup_{\ell} \mathcal{U}_{i\ell}|$ is the number of distinct institutions of work w_i . Authorships with no retained institution are excluded upstream, so $u_{i\ell} \geq 1$. Under author-fractional counting $\sum_u \omega_{iu} = 1$ (Waltman and van Eck, 2015). Institution pruning on annual output τ_U precedes fractional counting; this boosts a work’s remaining institutions but diminishes the work’s total contribution when some institutions are dropped.

We construct the set of *units* [sources or institutions; Pinski and Narin (1976)] $\mathfrak{P} = \mathfrak{S} \cup \mathfrak{I}$ with $N = N_s + N_u$ elements, ordered so that sources occupy indices $1, \dots, N_s$ and institutions indices $N_s + 1, \dots, N$. Each reference $r(i, j) \in \mathfrak{R}$ connects units associated with w_i to units associated with w_j , across four unit-type pairs: source–source (*SS*), source–institution (*SI*), institution–source (*IS*), and institution–institution (*II*).

For each work w_i we define the source membership one-hot vector $\mathbf{b}_i^S \in \mathbb{R}^{N_s}$ with $[\mathbf{b}_i^S]_s = \mathbf{1}[w_i \in s]$, and the institution weight distribution vector $\mathbf{b}_i^U \in \mathbb{R}^{N_u}$ with $[\mathbf{b}_i^U]_u = \omega_{iu}$ from (2). The vectors $\mathbf{b}$ yield the projection across unit pairs, accumulated as four block matrices over $\mathfrak{R}$ via rank-1 outer products,

$$\mathbf{C}_{XY} = \sum_{(i,j) \in \mathfrak{R}} \rho_i \mathbf{b}_i^X (\mathbf{b}_j^Y)^\top, \quad XY \in \{SS, SI, IS, II\}. \quad (3)$$

Here, $\mathbf{b}_i^X$ distributes the citing side across units of type X , and $\mathbf{b}_j^Y$ distributes the cited side across units of type Y . Each entry $[\mathbf{C}_{XY}]_{pq}$ is therefore proportional to the sum over $\mathfrak{R}$ of ρ_i -weighted fractional references from unit p of type X to unit q of type Y .

Convention. Throughout this paper, rows of $\mathbf{C}$ (and of its row-normalised form $\mathbf{H}$ below) index the citing side and columns index the cited side. Row sums are therefore references *given* and column sums are references *received*. This is the transpose of the convention used in parts of the economics-of-citations literature. We prefer “reference” to “citation” throughout; both are understood as the directed relation from citing to cited work.

The reference matrix assembles the four blocks:

$$\mathbf{C}(\chi, \mathbf{m}, \rho) = \begin{bmatrix} m_{SS}(1-\chi)^2 \mathbf{C}_{SS} & m_{SI}\chi(1-\chi) \mathbf{C}_{SI} \\ m_{IS}\chi(1-\chi) \mathbf{C}_{IS} & m_{II}\chi^2 \mathbf{C}_{II} \end{bmatrix}. \quad (4)$$

The binary *block mask* $\mathbf{m} = (m_{SS}, m_{SI}, m_{IS}, m_{II}) \in \{0, 1\}^4$ is bibliometrically meaningful for source-only (1000), institution-only (0001), bipartite (0110), and full-joint (1111) combined blocks. The mixing weight χ controls the source–institution balance in the bipartite and full-joint cases; in the single-mode cases, it enters only as an overall scaling and does not affect the ranking. The four coefficients form a partition of unity, $(1-\chi)^2 + 2\chi(1-\chi) + \chi^2 = 1$, so the full-joint mask distributes each work’s attention weight across block paths without inflating the total.

As a consequence of filtering $\mathfrak{R}$ to census-to-target references, the blocks $\mathbf{C}_{XY}$ exclude (a) census-work references not giving attention to target works, (b) non-census works giving attention to target works, and (c) authorships that include an omitted institution.

Ordinal spectral ranking. Let $r_p = \sum_q C_{pq}$ be the sum of fractionated references given by unit p and $\mathbf{D}_r = \text{diag}(r_p)$ the corresponding diagonal matrix. Row-normalising $\mathbf{C}$ gives the row-stochastic matrix $\mathbf{H} = \mathbf{D}_r^{-1} \mathbf{C}$, ensuring that, when $\rho_i = 1$, differences in reference intensity across units are removed. They are reintroduced later via the influence-per-work scaling (7). Dangling units (units with $r_p = 0$) and units with no incoming references are iteratively excluded from the strongly connected component on which ranking is performed, as described in the Computation paragraph below.

The three unipartite masks—source-only (1000), institution-only (0001), and full-joint (1111)—each yield a square $\mathbf{H}$ with the spectral ranking $\boldsymbol{\pi}^*$ (see Vigna, 2016, for an overview)

$$(\mathbf{I} - \mathbf{H}^\top)\boldsymbol{\pi}^* = \mathbf{0}, \quad \mathbf{1}^\top \boldsymbol{\pi}^* = 1, \quad \boldsymbol{\pi}^* \geq 0. \quad (5)$$

If $\mathbf{H}$ is primitive (irreducible and aperiodic) the ranking is unique by the Perron–Frobenius theorem. Our numerical implementation tests primitivity for every $\mathbf{H}$, pre-processed as explained below, and has not encountered an imprimitive case. Geller (1978) showed that $\boldsymbol{\pi}^*$ is the stationary distribution of the Markov chain with transition matrix $\mathbf{H}$, and that it coincides with the *influence weight* of Pinski and Narin (1976) scaled by the reference-count proportion $r_p / \sum_q r_q$.

Katz (1953) proposed attenuating walks (chains of reference-matrix weights $c_{ab} \rightarrow c_{bc} \rightarrow c_{cd}$) of length k by a factor α^k , $\alpha \in (0, 1)$, so that distant links count for less than proximate ones. Summing walks of all lengths and endowing every unit with a uniform initial score of 1 yields the Katz resolvent equation

$$(\mathbf{I} - \alpha \mathbf{H}^\top)\boldsymbol{\pi} = \mathbf{1}.$$

Because $\|\mathbf{H}\|_\infty \leq 1$, the spectral radius $\rho(\alpha \mathbf{H}^\top) < 1$ for any $\alpha \in (0, 1)$, so $(\mathbf{I} - \alpha \mathbf{H}^\top)$ is invertible by Neumann series regardless of dangling-row handling.

Hubbell (1965) generalised this by replacing the uniform endowment $\mathbf{1}$ with an informative prior $\boldsymbol{\mu}$ over units, representing exogenous information about standing independent of reference structure,

$$(\mathbf{I} - \alpha \mathbf{H}^\top)\boldsymbol{\pi} = \boldsymbol{\mu}.$$

We explore two priors, normalised so that $\mathbf{1}^\top \boldsymbol{\mu} = 1$:

Flat: $\mu_p = 1/N$ for every unit $p \in \mathfrak{P}$.

Type-balanced: each unit type carries equal total prior mass $1/2$, so $\mu_p = 1/(2N_s)$ for sources and $\mu_p = 1/(2N_u)$ for institutions.

The intermediate case $(\mathbf{I} - \alpha \mathbf{H}^\top)\boldsymbol{\pi} = \mathbf{0}$ with $\alpha \neq 1$ admits only the trivial solution $\boldsymbol{\pi} = \mathbf{0}$ and is not useful as a ranking.

Combining these yields the general (Katz–Hubbell) resolvent equation

$$(\mathbf{I} - \alpha \mathbf{H}^\top)\boldsymbol{\pi} = (1 - \alpha)\boldsymbol{\mu}, \quad \mathbf{1}^\top \boldsymbol{\pi} = 1, \quad \boldsymbol{\pi} \geq 0, \quad (6)$$

which recovers $\boldsymbol{\pi}^*$ in the limit $\alpha \rightarrow 1$ when $\mathbf{H}$ is primitive. The prior $\boldsymbol{\mu}$ has a substantive interpretation as exogenous standing independent of reference structure. Together with $\alpha < 1$ it additionally guarantees a unique nonnegative solution to (6) even when $\mathbf{H}$ is not primitive: $\alpha < 1$ makes $(\mathbf{I} - \alpha \mathbf{H}^\top)$ invertible, and a nonzero $\boldsymbol{\mu}$ lifts the equation off the homogeneous system. If $\mathbf{C}$ were a binary adjacency matrix with the flat prior, this would be the PageRank spectral ranking (Brin and Page, 1998).

The spectral ranking $\boldsymbol{\pi}$ is the steady-state probability vector of the Markov matrix $\mathbf{H}$, which we refer to as the probability distribution of attention or influence. The *influence per work* is computed as (Palacios-Huerta and Volij, 2004),

$$\mathbf{v} = A \mathbf{A}^{-1} \boldsymbol{\pi}, \quad \mathbf{A} = \text{diag}(a_p), \quad A = \mathbf{a}^\top \mathbf{1} = \sum_p a_p, \quad (7)$$

i.e. $v_p = (A/a_p) \pi_p$. Here, the unit work count $a_p = |\{i : w_i \in p\}|$ where p is a specific source or institution, and $\boldsymbol{\pi}$ denotes the solution to (6). The scalar A normalises $\mathbf{v}$ so that its a_p -weighted mean equals one: $\mathbf{a}^\top \mathbf{v} / A = \mathbf{1}^\top \boldsymbol{\pi} = 1$. A unit with $v_p = 1$ therefore has the corpus-average influence per work, and $v_p = 2$ indicates twice the average influence per work. Palacios-Huerta and Volij (2004) show that $\mathbf{v}$ is uniquely characterised by several reasonable axiomatic constraints (but see Serrano, 2004).

Bipartite ranking. With $\mathbf{m} = (0110)$, equation (4) yields block off-diagonal matrices $\mathbf{C}$ and $\mathbf{H}$, forming a directed bipartite network between sources (S) and institutions (I). This mask is used as the baseline model in Section 4. Partitioning rows and columns into the S and I blocks, the semi-stochastic matrix takes the form

$$\mathbf{H} = \begin{pmatrix} \mathbf{0} & \mathbf{H}_{SI} \\ \mathbf{H}_{IS} & \mathbf{0} \end{pmatrix}, \quad (8)$$

where $\mathbf{H}_{SI}$ connects sources to institutions and $\mathbf{H}_{IS}$ does the reverse. Because every path alternates $S \rightarrow I \rightarrow S \rightarrow \dots$, this structure makes $\mathbf{H}$ periodic with period 2.

Squaring $\mathbf{H}$ collapses two-step paths to separate unit types, giving one-unit relations

$$\mathbf{H}^2 = \begin{pmatrix} \mathbf{M}_S & \mathbf{0} \\ \mathbf{0} & \mathbf{M}_I \end{pmatrix}, \quad \mathbf{M}_S = \mathbf{H}_{SI}\mathbf{H}_{IS}, \quad \mathbf{M}_I = \mathbf{H}_{IS}\mathbf{H}_{SI}.$$

Both $\mathbf{M}_S$ and $\mathbf{M}_I$ are row semi-stochastic products of row semi-stochastic matrices.

We adopt the convention that a single SI or IS step is damped by $\sqrt{\alpha}$, so the round-trip $S \rightarrow I \rightarrow S$ incurs attenuation $\sqrt{\alpha} \cdot \sqrt{\alpha} = \alpha$, matching the one-step attenuation in the unipartite modes. This ensures the spectral radius of $\alpha\mathbf{M}_S$ equals that of the corresponding unipartite operator and makes α comparable across masks. Applying this substitution to (6), partitioning $\boldsymbol{\pi} = (\boldsymbol{\pi}_S, \boldsymbol{\pi}_I)$ and $\boldsymbol{\mu} = (\boldsymbol{\mu}_S, \boldsymbol{\mu}_I)$, and eliminating $\boldsymbol{\pi}_I$ yields the bipartite resolvent for sources,

$$(\mathbf{I} - \alpha\mathbf{M}_S^\top)\boldsymbol{\pi}_S = (1 - \sqrt{\alpha})(\boldsymbol{\mu}_S + \sqrt{\alpha}\mathbf{H}_{IS}^\top\boldsymbol{\mu}_I). \quad (9)$$

When $\alpha < 1$ and $\boldsymbol{\mu} \neq \mathbf{0}$, the right-hand side links institutions to sources via $\mathbf{H}_{IS}^\top$. The institution ranking is then recovered from the source ranking via the matrix–vector product

$$\boldsymbol{\pi}_I = \sqrt{\alpha}\mathbf{H}_{SI}^\top\boldsymbol{\pi}_S + (1 - \sqrt{\alpha})\boldsymbol{\mu}_I. \quad (10)$$

Computation. Defining the modified matrix

$$\tilde{\mathbf{H}} = \alpha\mathbf{H} + (1 - \alpha)\boldsymbol{\mu}\mathbf{1}^\top, \quad (11)$$

the resolvent $\boldsymbol{\pi} = \tilde{\mathbf{H}}^\top\boldsymbol{\pi}$ is equivalent to (6). This is the general form of our numerical problem, though most results reported below use $\alpha = 1$ and $\boldsymbol{\mu} = \mathbf{0}$, so $\tilde{\mathbf{H}} = \mathbf{H}$.

We construct many realisations of $\tilde{\mathbf{H}}$ to test unit pairings, subfield restrictions, other parameter choices, and for bootstrap uncertainty estimation. Each realisation entails a unit–unit reference network with one giant strongly connected component (SCC) together with a few, typically singleton, components that are disjoint from the main network. We retain the SCC and iteratively drop units whose row sum or column sum on the SCC is zero, alternating row and column drops until the residual $\tilde{\mathbf{H}}$ is strictly row-stochastic. Primitivity is then verified numerically as described below.

$\tilde{\mathbf{H}}$ is a sparse matrix constructed from the filtered and weighted edge list $\mathcal{E}$ accumulated over $\mathfrak{R}$, stored in compressed sparse row (CSR) format. We use two numerical approaches, applied to different regimes:

- For the baseline $\alpha = 1$, $\boldsymbol{\mu} = \mathbf{0}$ runs, the dominant eigenpair of $\tilde{\mathbf{H}}^\top$ is computed with `scipy.sparse.linalg.eigs`, which also returns the next few eigenvalues and thereby verifies primitivity (a unique eigenvalue of modulus 1).
- For $\alpha < 1$ runs (sensitivity and bootstrap), the power iteration

$$\boldsymbol{\pi}^{(k+1)} = \tilde{\mathbf{H}}^\top\boldsymbol{\pi}^{(k)} \quad (12)$$

is used. Because $(1 - \alpha)\boldsymbol{\mu}\mathbf{1}^\top$ makes $\tilde{\mathbf{H}}$ primitive, power iteration converges to $\boldsymbol{\pi}$ from any strictly positive starting vector $\boldsymbol{\pi}^{(0)}$.

In either approach, the per-work scaling (7) is applied after $\boldsymbol{\pi}$ has been computed.

3 Corpus selection

OpenAlex (Priem et al, 2022) provides our bibliometric data. Corpus selection starts with a list of eligible sources having three features: evidence from one of our journal registries that the source publishes economics or business research, an OpenAlex source identifier for corpus extraction, and a sufficient number of published works per year.

The journal registries are as follows. Harzing’s *71st Edition of the Journal Quality List* (Harzing, 2024, JQL) assembles peer assessments of journal quality across business and economics disciplines. Its scope is

confined to economics and business so no filtering is applied. The *ERA 2023 Submission Journal List* (Australian Research Council, 2022, SJL) assigns up to three Field of Research (FoR) codes to each entry. From the full list of 26,241 journals we retain those for which *every* assigned FoR code falls within Division 35 (Commerce, Management, Tourism and Services) or Division 38 (Economics)—specifically the four-digit groups 3501–3509 and 3801–3803, 3899. *Clarivate Web of Science Journal Citation Reports* (Clarivate Analytics, 2026, JCR) assign each journal a Web of Science category. We retain those in “Economics”, “Management”, “Business”, “Transportation”, or “Business, Finance”.

Every journal in JQL, SJL, and JCR has at least one ISSN, and may have several. These registry ISSNs are joined independently to the February 2026 OpenAlex sources snapshot using its list of ISSNs as the matching key and selecting only the ‘journal’ source-type. Around 30 additional matches are identified by exact title matching for OpenAlex sources that lack an ISSN. We denote the union of matched OpenAlex sources OAS^* .

For OAS^* sources we extract work and authorship data and examine the annual-output distributions for sources and institutions. Retention curves (Fig. 1) show that a majority of institutions and a substantial fraction of sources contributing fewer than 20 works per year in the census window, producing long tails that contribute little stable citation density. We therefore adopt the thresholds $\tau_S = 20$ works/source/year and $\tau_U = 20$ works/institution/year. Sources retained under τ_S form the set OAS ; institutions retained under τ_U form $\mathcal{U}$.

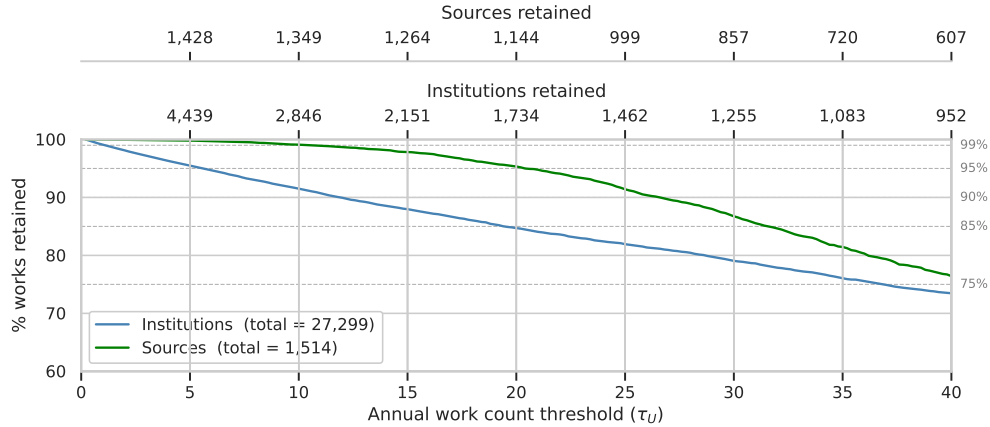


Figure 1: Institution and source retention curves.

Using sources and institutions that satisfy these thresholds, we extract works, authorships and references published between 2000 and 2024, retaining works typed as ‘article’ or ‘review’ and excluding works that are paratext, retracted, or without an authorship entity. For the baseline results of Section 4 the census and target windows coincide at $t^c = t^t = [2020, 2024]$; the full 2000–2024 extraction supports the time-series sensitivity runs $t_1, \dots, t_4$ reported there.

Sources span subfields F within the broad $\mathfrak{F}$ set of sources in Economics and Business, labelled as follows. Each source in OAS carries one or more subfield labels designating categories, groups, subjects, and topics in JQL, SJR, JCR, and OAJ respectively. These labels are mapped by hand to the broad subfields E (economics) and B (business), or left unmapped (e.g. Psychology, Education from JQL). Ambiguous labels (e.g. Finance, Accounting) are mapped to E . The mappings are counted, and the source classified as follows:

- ‘E’ when its E-count is ≥ 2 and its B-count is < 2 (economics-dominant);
- ‘B’ when its B-count is ≥ 2 and its E-count is < 2 (business-dominant);
- ‘A’ when both E-count and B-count are ≥ 2 (genuinely ambiguous);
- ‘X’ when both E-count and B-count are < 2 (weak signal in both fields).

Institutions generally publish in multiple subfields, so direct categorical classification is unsuitable. Instead, we characterise each institution by the stochastic reference matrix H_{SI} . For each source s , $H_{SI}[s, U]$ is

the fraction of s 's citation weight directed at institution u , so every source contributes equally regardless of its total output. The field- F score for institution u is $\sum_{k: \text{subfield}(k)=F} H_{SI}[k, u]$. An institution is assigned $F = X$ if its X -score is at least $\frac{1}{2}$; otherwise it takes the dominant label among $\{E, B, A\}$.

Table 1: Registry sources and their OpenAlex matches, classified by field. JQL = Harzing Journal Quality List; MJL = Web of Science Master Journal List; SJL = ERA Subject Journal List (FoR codes 35 and 38). A source appearing in multiple registries is counted in each. E/B/A/X are assigned by the combined four-signal scoring. OAS* is the union across all three registries matched to OpenAlex. OAS is OAS* restricted to sources with mean annual works $\geq \tau_S = 20$ in the census window 2020–2024.

Register	Sources	OA Match	Match%	E	B	A	X
JQL	842	835	99.2%	120	260	203	252
MJL	1,414	1,397	98.8%	317	460	371	249
SJL	1,603	1,380	86.1%	229	594	362	194
OAS*	2,294	—	—	395	770	446	683
OAS	1,688	—	—	282	557	335	514

4 Results

The model supports three structural arrangements: (1) the unit mask m picks out various unit blocks, (2) the subfield set $\mathcal{F}$ firstly how structural changes in the reference network (i) unit-related, (ii) field-related, (iii) time-related, (iv) parameter effects, and (v) bootstrap uncertainty estimates. Themes (i), (ii) and (iv) use the *baseline*, [2020–24, 2020–24], $F = A$, $\tau_U = \tau_S = 20$, $\rho = 0$, $\mathbf{m} = 0110$, $\alpha = 1$, $\mu = 0$ — the eigenvector equation (5) on the bipartite network.

Unit Figure 2 compares spectral ranks across network masks. The x -axis is the cardinal rank of sources (upper) and institutions (lower), with the solid black curve tracing the corresponding ordinal rank. The units near $v = 1$ lie at the weighted average influence-per-work, around rank position 25% for sources and 20% for institutions. Influence rises to > 10 for the top few units at the extreme left, with a low-influence tail falling below 0.1 for 10% of sources and $< 5\%$ for institutions. For sources, modes 1000 (red $\times$) and 1111 at χ^* (green $\times$) are overlaid on the bipartite baseline; open circles and squares on the 1000 series mark Economics and Business sources respectively. For institutions, mode 0001 and 1111 replace the source alternatives.

The most striking feature of Figure 2 is the contrast between the source and institution panels. For institutions, modes 0001 and 1111 track the bipartite baseline closely: prestige per work is largely insensitive to whether the institution-to-institution path is direct ($\mathbf{m} = 0001$) or mediated through sources ($\mathbf{m} = 0110$). For sources the picture is different: mode 1000 produces pronounced scatter both above and below the bipartite baseline, with a systematic downward shift for the majority and a distinct subset elevated well above it. The spectral gap $g = 1 - \lambda_2$ measures how quickly the random walk mixes across the network and thereby how strongly positional ranking depends on local neighbourhood rather than global structure. For the source-only mode $g^{SS} = 0.038$ —the walk on $\mathbf{H}_{SS}$ mixes very slowly, so a source's rank reflects its position within its local citation cluster rather than its standing in the full network. Routing prestige through the SI – IS blocks dissolves this locality: the bipartite mode achieves $g^{0110} = 0.660$, nearly seventeen times larger, because institutions draw citations from and distribute prestige to sources across the citation network. The institution-only mode sits at $g^{II} = 0.464$, already far above g^{SS} , reflecting the fact that institutions co-locate researchers who publish across many sources and therefore bridge the gaps that confine the source random walk. Mode 1111 combines both paths and achieves $g^{1111} = 0.347$.

The scatter in the $\mathbf{m} = 1000$ source panel is therefore a consequence of removing the SI – IS mediation that integrates the source network. Without institutional bridging, sources that happen to be the dominant nodes in their local citation neighbourhood are amplified in v relative to the bipartite baseline, while sources whose prestige in the bipartite ranking derives substantially from prestige flowing in via institutional co-authorship links are deflated. The institution panel of Figure 2 is smoother because the institution random walk never

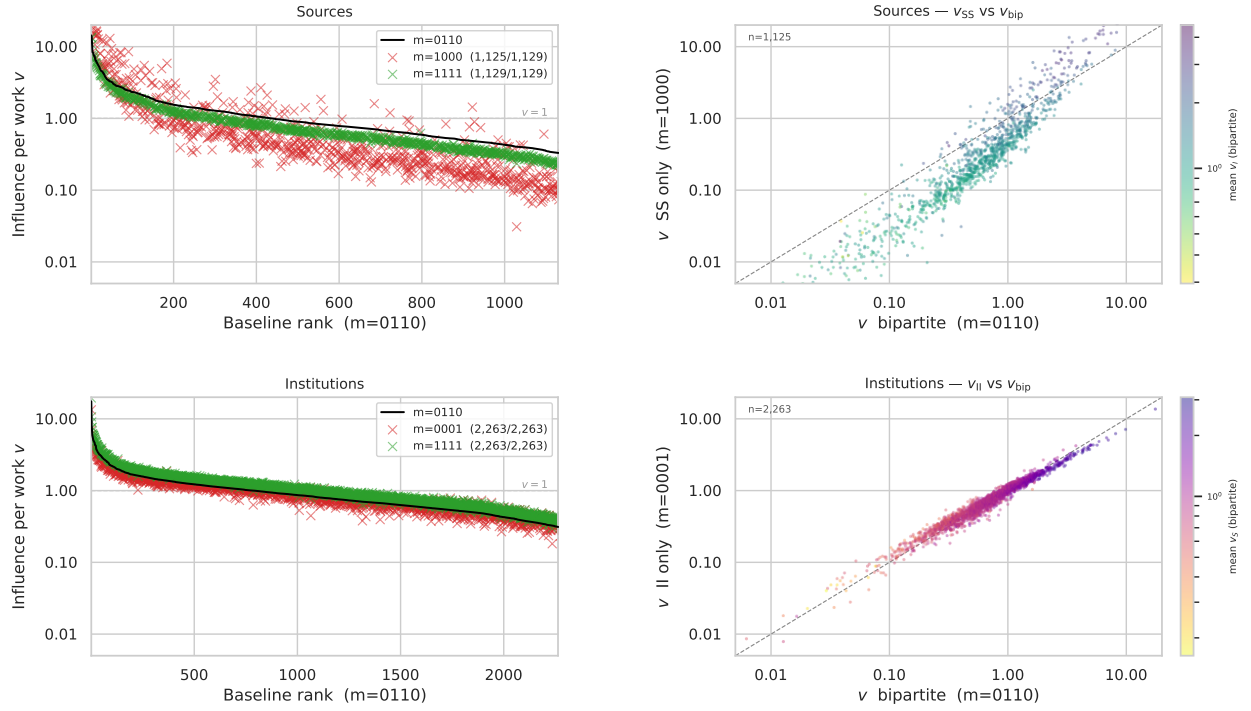


Figure 2: Prestige per work v against bipartite baseline rank. Top: sources. Bottom: institutions. Black line: bipartite baseline $\mathbf{m} = 0110$. Red $\times$: $\mathbf{m} = 1000$ (top) / $\mathbf{m} = 0001$ (bottom). Green $\times$: $\mathbf{m} = 1111$, $\chi^* = [FILL]$. Open circles ($\circ$) and squares ($\square$) overlaid on the 1000 series (top) and 0001 series (bottom) mark Economics and Business units respectively, classified by field-network membership. Parameters: $t_x = 5$, $F = A$, $\tau_U = \tau_S = 20$, $\rho = 0$, $\alpha = 1$.

relies on a narrow local structure to the same extent: the one-mode projection $\mathbf{M}_I = \mathbf{H}_{IS}\mathbf{H}_{SI}$ connects any two institutions that share sources cited by their authors, which creates a dense, well-connected graph even before the $IS-SI$ paths are included. This asymmetry between sources and institutions is a structural property of the bipartite network and not a feature of the particular corpus or time window.

The bipartite baseline is validated against external journal quality indicators in Figure 3, which plots the four JCR 2024 measures against v in log-log space for the 1,151 sources matched between the baseline ranking and the Web of Science Journal Citation Reports. All four indicators scale approximately as power laws in v : $\text{JIF}_{2024} \propto v^{0.55}$ ($R^2 = 0.52$), $\text{JIF}_{5\text{yr}} \propto v^{0.62}$ ($R^2 = 0.61$), $\text{JCI}_{2024} \propto v^{0.59}$ ($R^2 = 0.58$), and the Article Influence Score $\text{AIS} \propto v^{0.91}$ ($R^2 = 0.82$). AIS, which is itself an eigenvector-based measure constructed from the same citation flows, aligns most closely with v both in slope and goodness of fit. The sub-unity exponents mean that v is more compressed than any of the JCR indicators at the top of the distribution: the bipartite prestige-flow measure redistributes influence more evenly among high-ranking journals than raw citation counts do, consistent with the interpretation that a reference from a prestige-surplus source carries more weight than a reference from a low- v source regardless of volume. The correlation with AIS also confirms that the $SI-IS$ routing through institutions does not distort the basic signal; it refines it by incorporating the institutional dimension of research output.

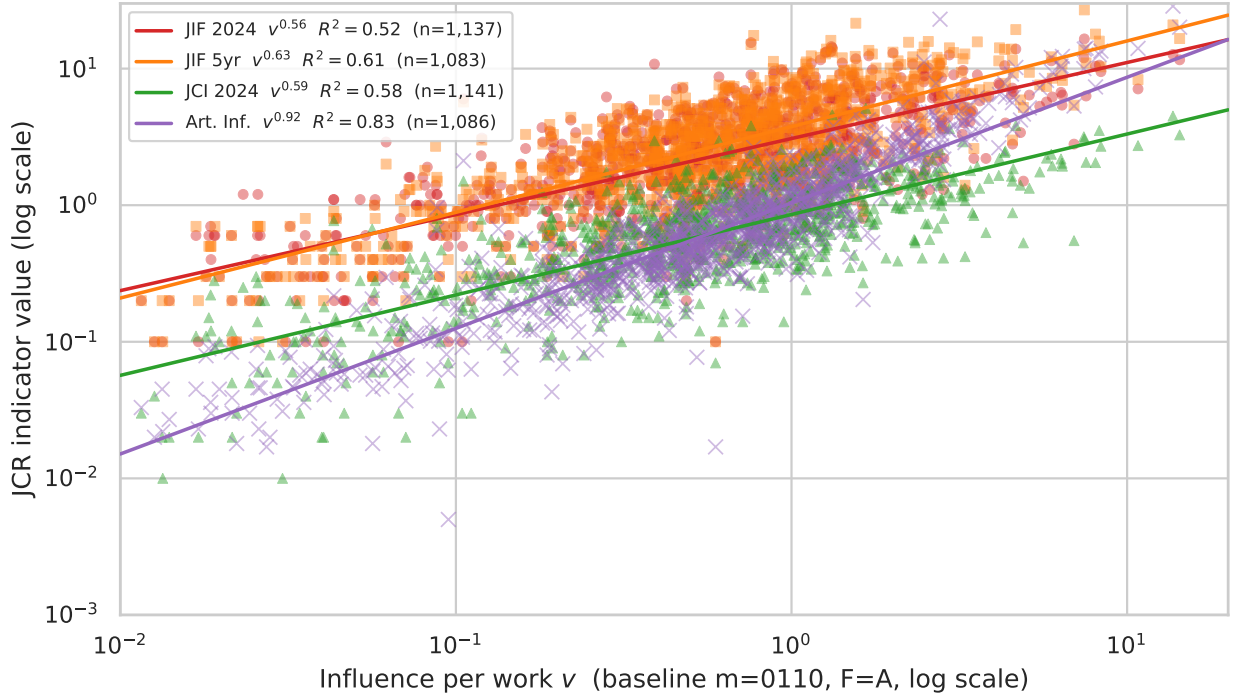


Figure 3: JCR 2024 indicators against prestige per work v (log-log). Scatter: individual sources matched to the Journal Citation Reports ($n = 1,151$). Lines: power-law OLS trend fitted in log-log space; legend reports exponent and R^2 . Dashed line: $v = 1$ (corpus average). Parameters: $t_x = 5$, $F = A$, $\tau_U = \tau_S = 20$, $\rho = 0$, $\mathbf{m} = 0110$, $\alpha = 1$.

Figure 4 plots v for all sources and institutions across the five time windows $t_x = 1, \dots, 5$ (2000–2024), with the x -axis locked to the baseline rank. The prestige hierarchy is visibly persistent: the dispersion around the baseline curve narrows as the window approaches the present, and even the most distant window ($t_x = 1$, 2000–04) reproduces the broad shape of the distribution. The Spearman correlation between v in 2000–04 and v in 2020–24 is $\rho_S = 0.663$ for sources and $\rho_S = 0.660$ for institutions on the common sets; for consecutive windows it rises to $\rho_S = 0.905$ (sources) and $\rho_S = 0.864$ (institutions), so the bulk of the apparent instability in Figure 4 is long-run drift rather than short-run volatility. Institutions are no less stable in rank than sources despite having more volatile absolute v values across windows, which reflects the growing size of the institution corpus as additional universities cross the τ_U threshold over the 25-year period.

The greater smoothness of the institution panel in Figure 2 and the comparable long-run rank stability in Figure 4 point to the same mechanism: the *SI-IS* coupling acts as a stabiliser, averaging out the local fluctuations in citation practice that would otherwise produce volatile source rankings.

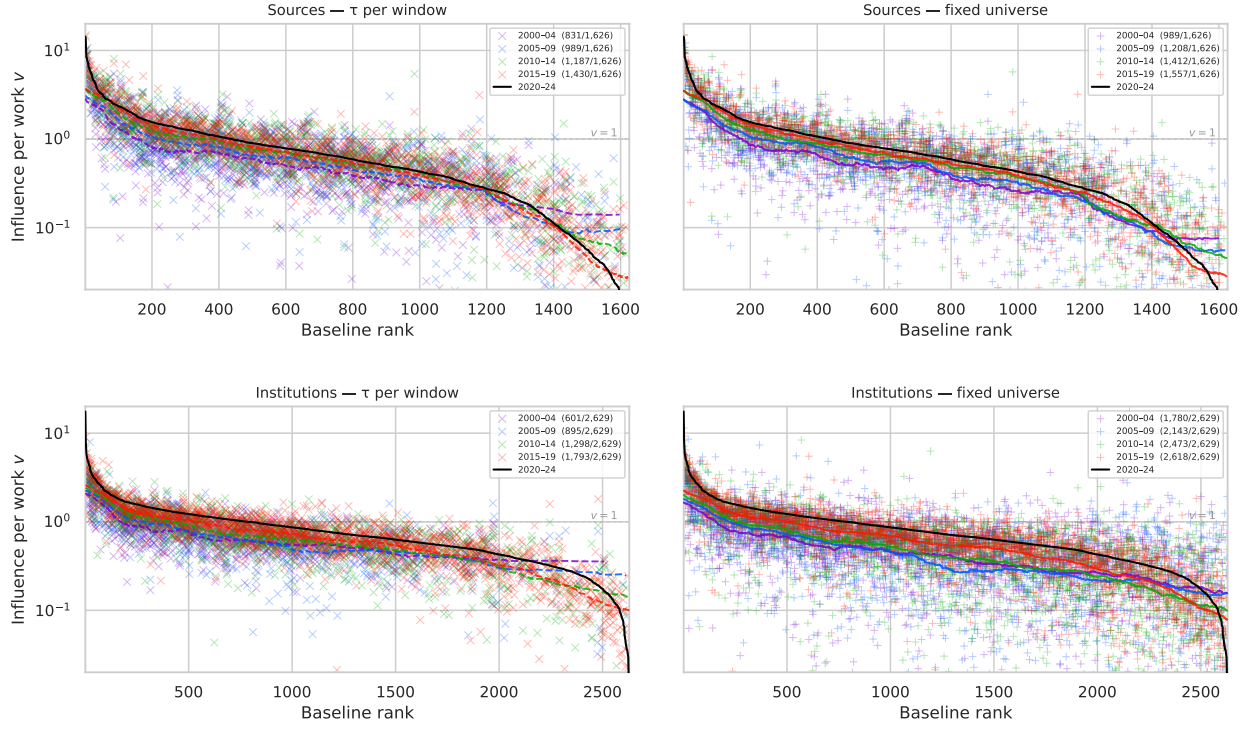


Figure 4: Prestige per work v against baseline rank across five time windows. Top: sources. Bottom: institutions. Black line: baseline $t_x = 5$ (2020–24). Coloured series: $t_x = 1$ (2000–04) through $t_x = 4$ (2015–19). x -axis locked to baseline rank; units absent from a given window are omitted from that series. Parameters: $F = A$, $\tau_U = \tau_S = 20$, $\rho = 0$, $\mathbf{m} = 0110$, $\alpha = 1$.

Figure 5 quantifies the persistence of *prestige-surplus* status—defined as $v > 1$, meaning the unit absorbs more prestige per work than the corpus average—across the full 25-year span. Approximately one quarter of sources and one quarter of institutions are prestige-surplus in each window, and this fraction is stable across time, confirming that $v = 1$ is a structurally consistent threshold rather than a feature of any particular window. Restricting to the 835 sources and 606 institutions present in both the 2000–04 and 2020–24 windows, 83% of sources and 94% of institutions that were prestige-surplus in 2000–04 retain that status in 2020–24. The scatter plots confirm this asymmetry visually: the upper-right quadrant (surplus in both windows) dominates, while the lower-right quadrant (surplus lost) is nearly empty. Classifying each unit’s trajectory as a binary sequence (H if $v > 1$, L otherwise) across the five windows shows that monotone paths dominate among institutions (101 monotone rises vs 29 oscillating), while for sources the counts are closer (74 monotone vs 80 oscillating). Once an institution crosses $v = 1$ it almost invariably stays above it; sources are somewhat noisier, but the direction is sustained. The institutions that crossed the threshold between 2000–04 and 2020–24 are predominantly universities in Western countries that expanded their economics and business research output substantially over the period, consistent with the growth of internationally oriented business schools and the associated increase in faculty size.

Phase 1: Sources, institutions and fields. Figure 6 shows the prestige-per-work distribution for the baseline and for the field subsets $F = E$ and $F = B$, with the x -axis locked to the baseline rank so that each unit sits at its baseline position and absent units are omitted from that series. Sources at the Economics–Business boundary lose prestige flow from one side under each restriction; the institution panel reflects the same effect through the authorship mapping.

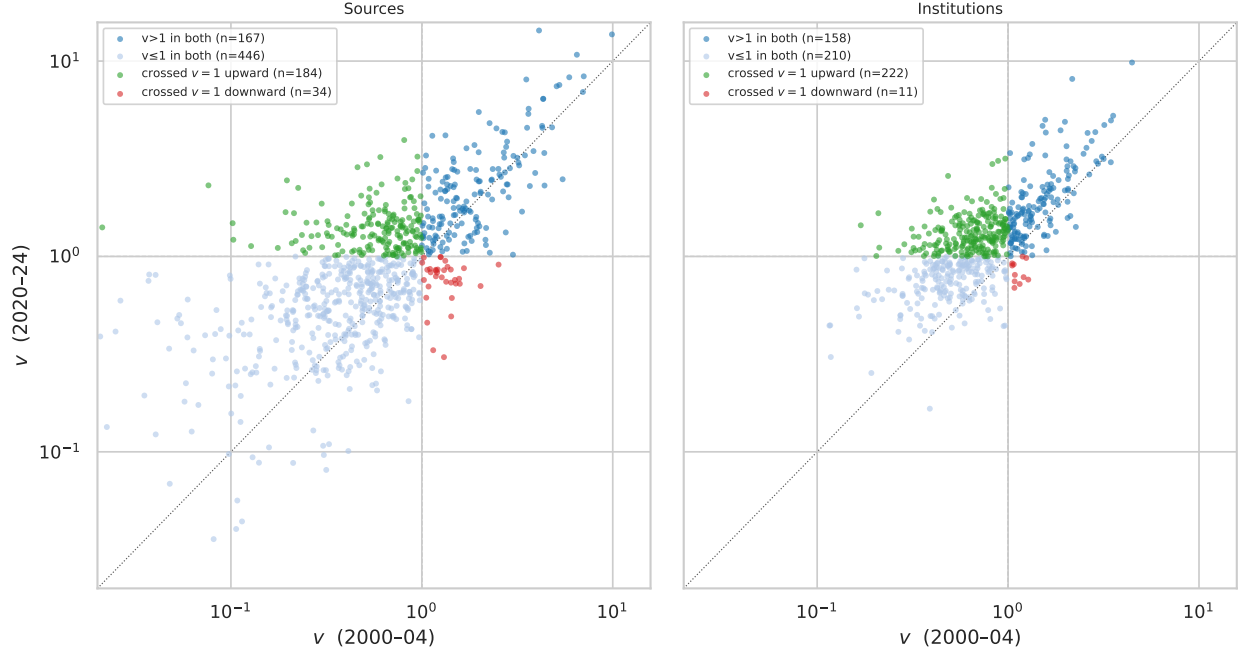


Figure 5: Prestige per work v in 2000–04 against v in 2020–24 for units present in both windows (log–log). Left: sources ($n = 835$). Right: institutions ($n = 606$). Colour indicates threshold-crossing status: blue = prestige-surplus ($v > 1$) in both windows; light blue = below threshold in both; green = crossed $v = 1$ upward; red = crossed $v = 1$ downward. Dotted diagonal: $v_{t_1} = v_{t_5}$. Dashed lines: $v = 1$ on each axis. Parameters: $F = A$, $\tau_U = \tau_S = 20$, $\rho = 0$, $\mathbf{m} = 0110$, $\alpha = 1$.

The field shifts in Figure 6 reflect community structure in the citation network: sources cluster into Economics and Business communities that preferentially cite within their own field, and this influences rankings because prestige propagates more readily within a community than across it.

The second eigenpair (λ_2, ϕ_2) of $\mathbf{H}^\top$ characterises this structure precisely. For general α and uniform prior $\mu_p = 1/N$, the solution to (6) has the spectral expansion

$$\boldsymbol{\pi} = (1 - \alpha) \sum_{i \geq 1} \frac{\phi_i^\top \boldsymbol{\mu}}{1 - \alpha \lambda_i} u_i, \quad (13)$$

where (λ_i, u_i, ϕ_i) are the eigenvalues, right eigenvectors, and left eigenvectors of $\mathbf{H}^\top$, ordered $\lambda_1 \geq \lambda_2 \geq \dots$. The $i = 1$ term recovers $\boldsymbol{\pi}^*$ as $\alpha \rightarrow 1$; the $i = 2$ term is the leading community correction, amplified by $1/(1 - \alpha \lambda_2)$ relative to all higher terms. The spectral gap $g = 1 - \lambda_2$ therefore governs community influence: small g means community membership strongly shapes rankings; large g means rankings are community-neutral. For the SS mode ϕ_2 is expected to separate Economics from Business sources; for II the partition is more diffuse because institutions co-locate faculty from both fields. In the bipartite mode the community operator is the one-mode projection $\mathbf{M}_S = \mathbf{H}_{SI} \mathbf{H}_{IS}$; ϕ_2^S is its second eigenvector and the institution community vector is recovered as $\phi_2^I = \mathbf{H}_{SI}^\top \phi_2^S$ (unit-normalised). Throughout, ϕ_2 is signed so that the unit with the largest work count a_p has $\phi_{2,p} > 0$.

Figure 7 illustrates these eigenvectors by plotting the size-adjusted community measure $\phi_{2,p}/\sqrt{v_p}$ against v_p ; dividing by $\sqrt{v_p}$ removes the normalisation artefact that suppresses $|\phi_{2,p}|$ for low-prestige units.

In the SS source panel (top left), the sign of $\phi_{2,p}/\sqrt{v_p}$ [FILL: cleanly separates Economics (positive) from Business (negative) journals across the full prestige range, with Finance journals concentrated near zero, confirming their role as citation bridges between the two communities]. The bipartite source panel (top right) shows [FILL: a similar partition / greater mixing], with spectral gaps $g^{SS} = [\text{FILL}]$ and $g^{\text{bip}} = [\text{FILL}]$ indicating that routing citation flow through institutions [FILL: partially dissolves / preserves] the community

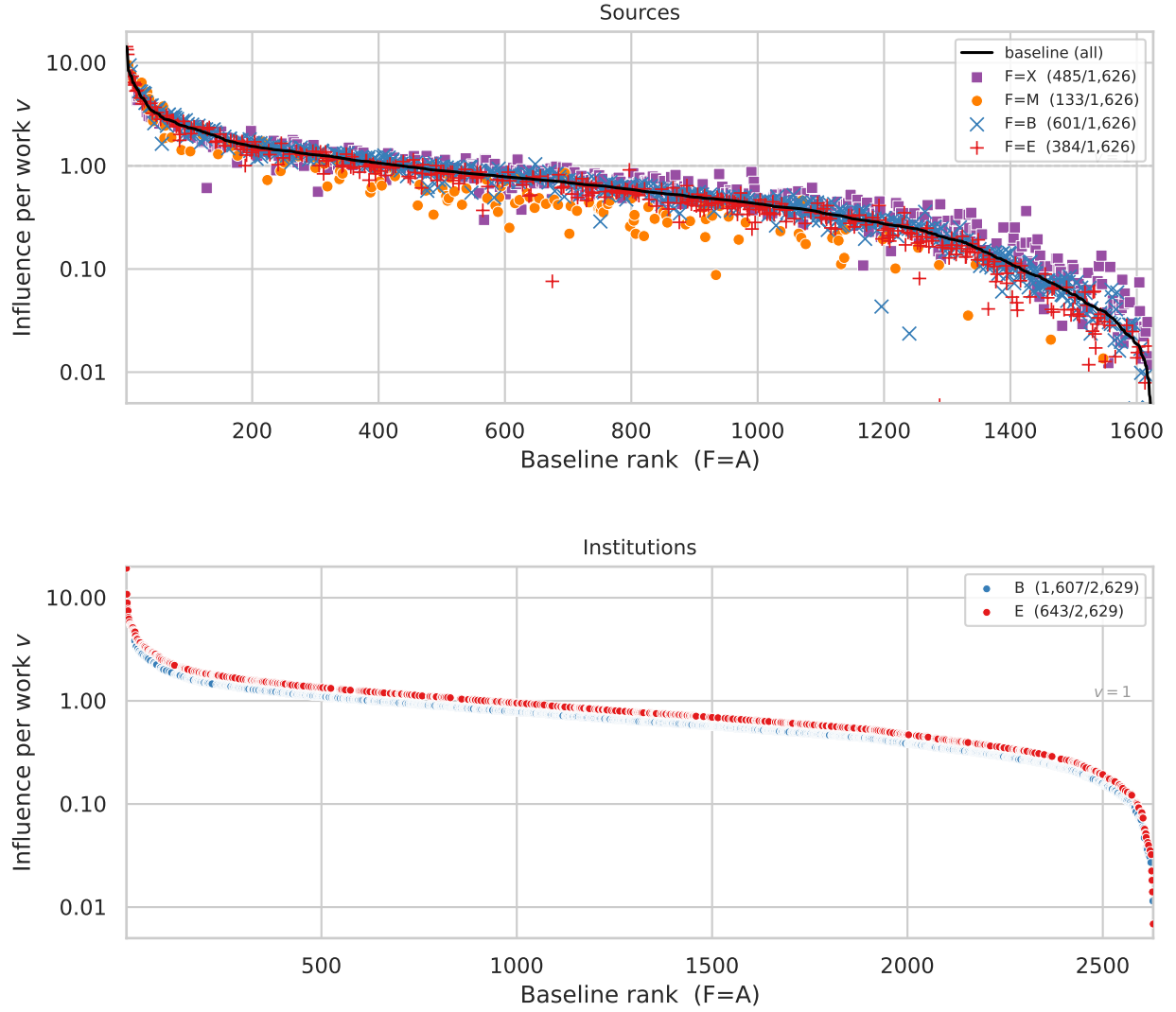


Figure 6: Prestige per work v against baseline rank. Top: sources. Bottom: institutions. Black line: baseline ($F = A$). Red +: $F = E$ ($N_s = [FILL]$). Blue $\times$: $F = B$ ($N_s = [FILL]$). Dashed line: $v = 1$ (corpus average). Parameters: $t_x = 5$, $\tau_U = \tau_S = 20$, $\rho = 0$, $\mathbf{m} = 0110$, $\alpha = 1$.

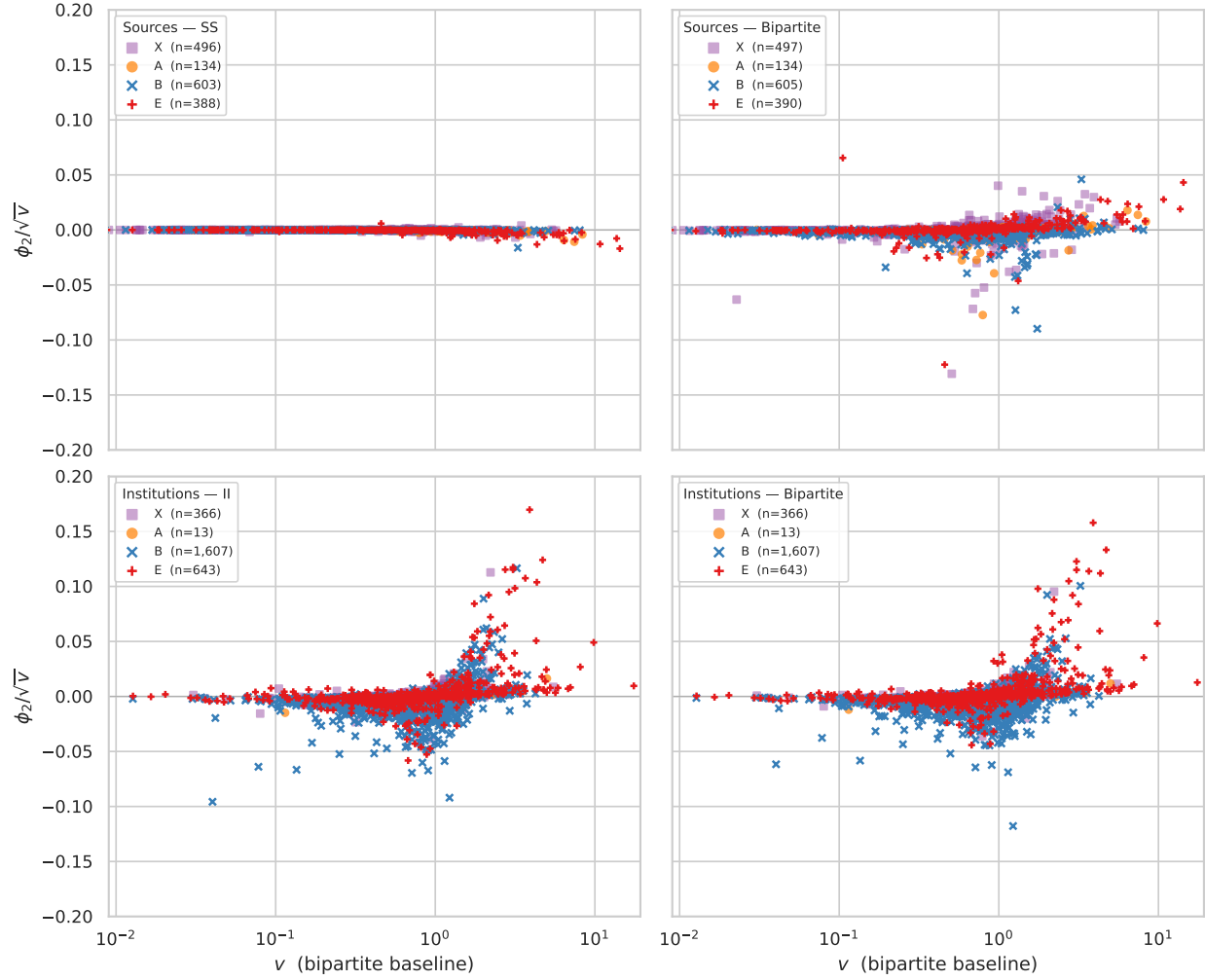


Figure 7: Community measure $\phi_{2,p}/\sqrt{v_p}$ against prestige per work v_p . Top row: sources. Bottom row: institutions. Left column: SS mode $\mathbf{m} = 1000$. Right column: bipartite $\mathbf{m} = 0110$. Sources coloured by field: blue circles = Economics (E); red squares = Business (B); grey crosses = other. Institutions coloured by field-network membership (E-only, B-only, other). x -axis: bipartite baseline v . Parameters: $t_x = 5$, $F = A$, $\tau_U = \tau_S = 20$, $\rho = 0$, $\alpha = 1$.

boundary. The institution panels (bottom row) show a more diffuse partition, consistent with institutions co-locating faculty from both fields; the field-network colouring confirms that E-only institutions [FILL: concentrate on one side / are scattered].

5 Discussion

Face validity. [Placeholder.]

Content validity. [Placeholder.]

Criterion validity — concurrent and predictive. [Placeholder.]

Construct validity. [Placeholder.]

6 Prospects

[Placeholder.]

7 Addenda

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